

CS 112

Introduction to Data Structures

WEEK 13

Parallelism: Moore's law, fork- join, OpenMP

Eric Araújo

Calvin University · Fall 2026

This Week

- 1 Moore's law
- 2 Why clock speed stopped
- 3 The fork-join paradigm
- 4 `#pragma omp parallel for`
- 5 Memory access patterns

Deck not yet written

This deck has NOT been written yet. Its content comes from [archive/26sp-slides/CS112-Multiprocessing.pptx](#) and Joel Adams' matrix lab ([archive/26sp-cs112/content/week12/matrix-mult-lab.md](#)).

Week 13 Recap

- `set` and `map` are balanced trees with a friendly interface
- Both keep keys **sorted**; both are $O(\lg n)$ per operation
- A set holds each value once; a map holds each key once
- `m[k]` **inserts** when the key is missing: use `count` to ask
- Need order? tree. Need raw speed? hash table

Next: a15 →

This deck stands on earlier CS112 materials by
Joel Adams and **Victor Norman** · adapted and extended by **Eric Araújo**

